Xiao Fu

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Education

Zhejiang University Hangzhou, China
Bachelor of Information Engineering | College of Information Science&Electronic Engineering (ISEE) Sept. 2018 – Jul. 2022

- **GPA:** 3.95/4.00, 90.53/100, **Ranking:** 4/145 (Top 3%)
- **Two-time National Scholarship winner, Graduated with Dean's Honor**
- **Honors Program:** Zhejiang University Morningside Cultural China Scholars Program (36 students selected from 12,000 freshmen with distinguished innovation capability and leadership)

Research Interests

3D Computer Vision, Neural Rendering and Scene Representation, 3D Reconstruction, Convex Optimization

Publications

∗ **equal contribution, † corresponding author**

- Xiao Fu*, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao†. "**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**". In *Proceedings of the International Conference on 3D Vision (3DV)*, 2022 [**Project Page**|**Paper**|**Code**]
- Guangyi Zhang*, Xiao Fu*, Qiyu Hu†, Yunlong Cai, Guanding Yu. "**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**". *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [**Paper**|**Code**]

Research Experience

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer Max Planck Institute for Intelligent Systems
Mentor: Prof. Andreas Geiger and Prof. Yiyi Liao *Sept. 2021 – Mar. 2022*

- proposed to utilize neural rendering as a tool to generate high-quality 2D panoptic segmentation annotations from a set of sparse images and easy-to-obtain coarse 3D bounding primitives and noisy 2D semantic predictions
- leveraged a novel dual formulation of the semantic fieds to improve the underlying geometry reconstruction and resolve label ambiguity at the intersection region of the 3D bounding primitives
- enabled rendering multi-view and spatio-temporally consistent labels at novel viewpoints

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network Zhejiang University
Mentor: Prof. Guanding Yu *Jun. 2020 – Feb. 2021*

- proposed a learning-based framework to address the high computational complexity problem of a convex optimization algorithm, i.e. the dual-layer iterative Penality Dual Decomposition (PDD) algorithm
- unfolded the PDD algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

Honors & Awards

- National Scholarship, Ministry of Education of P.R. China **(Top 0.2% across China)** *2020,2021*
- ISEE Excellent Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Dean's Award for Academic Contest, Bonus 20,000 CNY)** *2020,2021*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy **(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Outstanding Graduation Thesis Award of Zhejiang University *2022*
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Gold Prize, 7^{th} "Inernet Plus" College Student Innovation and Entrepreneurship Contest (1^{st} **author)** *2021*
- Gold Prize, 12^{th} "Challenge Cup" National College Student Business Plan Competition, National *2020*
- Grand Prize, "Yongdian Cup" Innovation and Entrepreneurship Contest (1^{st} **author)** *2019*
- Second Prize, College Student Physical Innovation Competition (Theoretical) of Zhejiang Province *2020*
- Third Prize, 17^{th} "Challenge Cup" Extracurricular Academic Science and Technology Contest (1^{st} **author)** *2021*
- Outstanding Student Award of Zhejiang University *2019,2020,2021*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*